

End-to-End Secondary-Structure Prediction Pipeline

A. Model Methodology & Architecture

Our final pipeline uses a Bidirectional Long Short-Term Memory (BiLSTM) network with Position-Specific Scoring Matrix (PSSM) features. We chose this deep learning method over classical machine learning models, like Support Vector Machines or Random Forests using simple one-hot encoding. During the Week 5 validation phase, we saw that evolutionary information from PSSMs gives the model extra context. PSSMs represent the frequency of amino acid substitutions at each position across evolutionarily related proteins. This helps the model resolve long-range dependencies in the sequence, and it performed better than models that only looked at the raw amino acid sequence.

The network architecture is set up in a few specific layers. The input layer takes a 21-character amino acid vocabulary and maps it to an embedding dimension of 32. Next, the feature concatenation step takes these learned embeddings and combines them with the 20-dimensional PSSM features for each individual residue. For the recurrent layers, we use a BiLSTM with a hidden size of 64. This bidirectional approach is useful because it processes the sequence from left to right and right to left, allowing the model to capture both upstream and downstream context for any given residue. Finally, the output head consists of a linear layer that converts the 128-dimensional output, which comes from 64 dimensions in each direction, into the 3-state Q3 classification of H, E, and C.

We trained the model using Cross-Entropy loss alongside basic regularization techniques. We also maintained strict homology

separation during the train, validation, and test splits. A fixed random seed of 42 was used to generate these data splits before any inference took place, ensuring there was absolutely no test-set leakage. The saved artifacts, which include `bilstm_pssm.pt`, `config.json`, `split.json`, and `manifest.json`, contain the frozen weights, the hyperparameters, and the exact split protocols. These files are required to reproduce the predictions without having to retrain the network from scratch.

B. Results & Error Analysis

On the held-out 10-chain test set, the frozen model achieved a Q3 accuracy of 61.96% for the raw predictions and 60.22% for the smoothed predictions. We used a `min_segment=3` smoothing function, but it actually lowered the overall accuracy. This happened because the smoothing process erased several correct short-strand predictions by forcing them to match the surrounding states.

Looking at the 3x3 confusion matrix, we can break down the performance by class. The recall for the H class was 75.2%. The C class recall was 67.8%. The E class recall was much lower at 23.0%. The β -strand (E) class was clearly the hardest secondary structure for the model to predict. Strands physically depend on long-range hydrogen bonding with distant residues in the folded protein. These interacting residues might be very far apart in the primary sequence. Because our model looks at local window context along a 1D sequence, it struggles to identify these distant relationships. As a result, the model misclassified E states fairly evenly into both H and C states. Specifically, true E residues were predicted as H 151 times and as C 124 times.

We also looked at accuracy at segment boundaries compared to the interior of segments. The accuracy is much lower at segment boundaries. Transitional residues usually have less clear structural properties

because they sit between two distinct folded states. When we visualized accuracy compared to segment length, the plot showed that longer segments are predicted more accurately than short ones. The model has an easier time recognizing stable, established structures compared to short or fluctuating runs.

For the per-protein diagnosis, protein 2OOK_A stood out as an extreme outlier. It scored only 33.9% accuracy, which is 34 points below the next worst protein. We analyzed the sequence composition and found that 2OOK_A is unusually β -rich. It consists of 37.8% E states, compared to the 19.4% average across the rest of the test set. Since the model already has a known weakness at predicting the E-class, its performance dropped significantly on this β -heavy protein.

To try and improve the poor E-class recall, we trained a new model iteration. For this version, we used a class-weighted loss function. The goal was to penalize the model more heavily when it made errors on the E-class. The result of this iteration was that E recall went up from 23.0% to 27.7%. The E precision dropped from 51.2% to 39.3%. This trade-off brought the overall raw accuracy down to 59.78%. The model essentially traded false negatives for false positives. It started guessing the E state more often instead of actually learning the underlying patterns of the class. This experiment confirmed that the baseline unweighted model is still better for overall generalization.

C. Biological Inference

We looked at the biological meaning of our predictions for three specific test chains to see what the model outputs imply about protein function.

First, we examined 2RG8_A, which was identified as a transmembrane candidate. The model predictions for this protein show a ≥ 20 consecutive H-run. Overall, the sequence has a 68% helix composition and only about 1% strand composition. This structural breakdown points to an all- α structure. Because of the long unbroken helix prediction, it likely functions as a single-pass transmembrane helix or possibly a helical bundle. It does not have the alternating pattern required for a 7-TM GPCR.

Second, we analyzed 1OCY_A, which was flagged for an Intrinsically Disordered Region (IDR). This chain contained a continuous coil run of 69 residues. This is well over the >30 residue threshold typically used to identify IDRs. The protein was also 77% coil overall. This high coil content indicates an Intrinsically Disordered Region, meaning the protein probably has a flexible structure rather than folding into a rigid globular shape.

Third, we did a functional cross-check on 2QSK_A. This sequence contains a very high number of cysteine residues. We checked the UniProt annotations and the AlphaFold DSSP records for this protein. The databases show that the model correctly found constrained loop and coil regions. These regions are structurally restricted by localized disulfide bonds forming between the cysteines.

During this analysis, we also noticed a systematic bias in the model. It frequently over-predicts the length of unbroken H-runs. For example, the model predicted a maximum H-run of 40 residues for the protein 2OOK_A. The true label for that run was only 15 residues long. Because of this tendency to extend helices, any predicted TM-helix counts from this model would need to be validated experimentally before drawing firm biological conclusions.

D. Limitations & Where Next

There are a few noticeable limitations and failures in our current pipeline. One major issue showed up when we were parsing the test chain 4FHR_B. The sequence was truncated and missing its final 27 residues. This happened because of a querying error when running PSI-BLAST. The BiLSTM model still ran successfully on the 189 available PSSM vectors, but it shows that the pipeline depends heavily on having perfectly generated PSSM data upstream. If the input alignment fails, the prediction is cut short. Also, relying strictly on 1D sequence data limits our ability to achieve high β -strand recall. 3D spatial proximity and long-range interactions are very hard to infer just from reading a sequence from left to right.

Looking forward, secondary structure prediction is mostly a solved problem in modern computational biology. The focus of the field is moving toward full 3D tertiary and quaternary structure modeling, such as AlphaFold 3. Researchers are also focusing on direct function prediction and mapping the effects of genetic variants. Still, the basic methods we used in this assignment are relevant. Using evolutionary alignments like PSSMs and MSAs, along with bidirectional neural networks to read sequence context, are important concepts. These same principles are built into the larger models that generate protein structures today.